Diskussionspunkte fürs nächste Meeting

1. Very low kWhDelivered compared to duration of charging sessions:

```
sessions[(sessions.kWhDelivered < sessions.kWhDelivered.quantile(0.01)) &
        (sessions.chargingTime > sessions.chargingTime.quantile(0.95)) ]
```

	connectionTime	disconnectTime	doneChargingTime	kWhDelivered	siteID	spaceID	blockedTime	chargingTime
6075	2019-07-15 05:57:53-07:00	2019-07-15 16:31:45-07:00	2019-07-15 15:54:20-07:00	0.727	1	AG-3F25	0 days 10:33:52	0 days 09:56:27
14009	2019-12-23 06:21:52-08:00	2019-12-23 15:38:10-08:00	2019-12-23 15:33:56-08:00	0.548	1	AG-1F04	0 days 09:16:18	0 days 09:12:04
58147	2019-02-11 06:58:52-08:00	2019-02-11 18:33:52-08:00	2019-02-11 17:38:49-08:00	0.731	1	AG-1F11	0 days 11:35:00	0 days 10:39:57
58813	2019-02-26 04:47:44-08:00	2019-02-26 15:31:18-08:00	2019-02-26 14:36:19-08:00	0.727	1	AG-1F14	0 days 10:43:34	0 days 09:48:35
64168	2019-06-13 07:14:38-07:00	2019-06-13 17:10:39-07:00	2019-06-13 17:00:34-07:00	0.573	1	AG-4F51	0 days 09:56:01	0 days 09:45:56

2. Some EVSE at site 2 showcase very low utilization (total sessions) with extended gaps with 0% utilization→stations may have been inop or new ones added:

```
print(sessions[(sessions.siteID==2) ].spaceID.value_counts().head(6),
      sessions[(sessions.siteID==2) ].spaceID.value_counts().tail(6))
```

```
CA-303    1696
CA-305    1330
CA-307    1061
CA-315    1024
CA-311     989
CA-309     970
Name: spaceID, dtype: int64 CA-509    81
CA-505    73
CA-149    72
CA-213    14
CA-148     7
CA-496     4
Name: spaceID, dtype: int64
```

```
sessions[(sessions.siteID==2) & (sessions.spaceID=='CA-496') ]
```

	connectionTime	disconnectTime	doneChargingTime	kWhDelivered	siteID	spaceID	blockedTime	chargingTime
27089	2019-08-13 10:04:48-07:00	2019-08-13 18:36:52-07:00	2019-08-13 11:43:47-07:00	13.076000	2	CA-496	0 days 08:32:04	0 days 01:38:59
38386	2018-08-22 11:48:07-07:00	2018-08-22 21:02:23-07:00	2018-08-22 20:21:45-07:00	13.002000	2	CA-496	0 days 09:14:16	0 days 08:33:38
42604	2018-10-15 08:13:17-07:00	2018-10-15 22:41:53-07:00	2018-10-15 19:59:31-07:00	15.039438	2	CA-496	0 days 14:28:36	0 days 11:46:14
44037	2018-11-03 15:57:44-07:00	2018-11-03 19:04:15-07:00	2018-11-03 18:05:14-07:00	1.601431	2	CA-496	0 days 03:06:31	0 days 02:07:30

```
sessions[(sessions.siteID==2) & (sessions.spaceID=='CA-148') ]
```

	connectionTime	disconnectTime	doneChargingTime	kWhDelivered	siteID	spaceID	blockedTime	chargingTime
24152	2019-05-02 12:37:18-07:00	2019-05-02 14:45:40-07:00	2019-05-02 14:44:09-07:00	0.768	2	CA-148	0 days 02:08:22	0 days 02:06:51
24659	2019-05-18 13:50:50-07:00	2019-05-18 17:41:46-07:00	2019-05-18 14:21:34-07:00	0.907	2	CA-148	0 days 03:50:56	0 days 00:30:44
39266	2018-09-03 11:05:27-07:00	2018-09-03 12:37:13-07:00	2018-09-03 12:37:11-07:00	10.184	2	CA-148	0 days 01:31:46	0 days 01:31:44
39426	2018-09-05 11:11:10-07:00	2018-09-05 11:43:05-07:00	2018-09-05 11:43:01-07:00	1.636	2	CA-148	0 days 00:31:55	0 days 00:31:51
39805	2018-09-10 09:22:56-07:00	2018-09-10 14:59:04-07:00	2018-09-10 10:09:37-07:00	1.373	2	CA-148	0 days 05:36:08	0 days 00:46:41
42876	2018-10-18 13:15:37-07:00	2018-10-18 16:53:38-07:00	2018-10-18 15:50:40-07:00	5.450	2	CA-148	0 days 03:38:01	0 days 02:35:03
47193	2021-05-19 11:47:01-07:00	2021-05-19 12:32:40-07:00	2021-05-19 12:17:36-07:00	0.806	2	CA-148	0 days 00:45:39	0 days 00:30:35

Problematic for prediction utilization, as there may have been fewer than 52 (55) stations operative in the past

3. Site 1 → majority of customers use the app
   Site 2 → only 50% of customers use the app
   Consider user data for site 1 and not for site 2? If at all…